

Nowell Hollier Phelps

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Education

PhD, Department of Biostatistics, Johns Hopkins University 2024 – 2029 (expected)

- Coursework in statistical theory, measure-theoretic probability, biostatistical methods, and statistical computing
- Advised by Dr Abhirup Datta

MMath, Mathematics University of Oxford 2017 - 2021

- Part C: Distinction; Parts A&B: Second Class, Division One
- Dissertation “A coevolutionary voter model for opinion dynamics on hypergraphs”, advised by Dr Renaud Lambiotte.

Research Experience

Visiting Researcher African Institute for Mathematical Sciences Research and Innovation Centre March - July 2024
Research visit at AIMS RIC in Kigali, Rwanda, collaborating on developing a database for climate and health research.

Research Assistant in Epidemiology and Population Health Imperial College London March 2022 - July 2024
RA in the NCD Risk Factor Collaboration, WHO Collaborating Centre on NCD Surveillance, Epidemiology and Modelling, supervised by Dr Majid Ezzati. Led analysis updating the WHO’s national underweight and obesity prevalence estimates.

Summer Studentship Big Data Institute, University of Oxford July - September 2021
Eight-week research internship investigating spatial risk of schistosomiasis, supervised by Dr Goylette Chami.

Undergraduate Research Internship Mathematical Institute, University of Oxford August - September 2020
Four-week research internship on social network effects on microfinance adoption, supervised by Dr Neave O’Clery.

Publications

1. **[Co-first author]** NCD Risk Factor Collaboration (NCD-RisC). “Obesity rise plateaus in developed nations and accelerates in developing nations”, *Nature* (2026)
2. NCD Risk Factor Collaboration (NCD-RisC). “Worldwide trends in diabetes prevalence and treatment from 1990 to 2022: a pooled analysis of 1108 population-representative studies with 141 million participants”, *The Lancet* (2024)
3. NCD Risk Factor Collaboration (NCD-RisC). “General and abdominal adiposity and hypertension in eight world regions: a pooled analysis of 837 population-based studies with 7.5 million participants”, *The Lancet* (2024)
4. **[Co-first author]** NCD Risk Factor Collaboration (NCD-RisC) “Worldwide trends in the burden of underweight and obesity from 1990 to 2022: a pooled analysis of 3663 population-representative studies with 222 million school-aged children, adolescents and adults”, *The Lancet* (2024)
5. NCD Risk Factor Collaboration (NCD-RisC). “Global variation in diabetes diagnosis and prevalence based on fasting glucose and hemoglobin A1c”, *Nature Medicine* (2023)
6. NCD Risk Factor Collaboration (NCD-RisC). “Diminishing benefits of urban living for children and adolescents’ growth and development”, *Nature* (2023)

Teaching Experience

Teaching Assistant, Department of Biostatistics, Johns Hopkins University
Advanced Probability Theory I and II August - December 2025

- PhD-level classes on measure-theoretic probability. Weekly office hours; grading; class on sequences taught 4 September 2025; review class taught 15 October 2025. Provided extra support in linear algebra foundations.

Advanced Methods in Biostatistics III and IV January - May 2026

- PhD-level classes on biostatistical methods. Weekly office hours; bi-weekly lab session; grading.

Tutoring for departmental qualifying exam May - June 2026

Tutoring for first-year PhD student on *Advanced Methods in Biostatistics I-IV* qualifying exam

Presentations and Posters

Invited presentations and panels

1. “The Obesity Pandemic: Trends and Perspectives” *Webinar: Addressing the Obesity Pandemic, Zurich Heart House, Foundation for Cardiovascular Research* (April 30th, 2026)
2. “Understanding the Global Burden of Obesity and the Implications for Cardiovascular Health” *Lancet panel on obesity and cardiovascular health outcomes, American Heart Association Scientific Sessions* (November 26th, 2024)
3. “MCMC for a Large Bayesian Hierarchical Model” *Statistics Seminar Group, University of Kent* (October 12th, 2023)

Conference posters

1. “Outdoor Air Pollutant Exposure is Associated with Accelerated Lung Function Decline in a High-Risk HIV Cohort” *American Thoracic Society International Conference* (May 20th, 2026)

Departmental presentations

1. “Modelling Trends in Population-Level Obesity” *Biostatistics Education and Student Training Seminar, Department of Biostatistics, Johns Hopkins University* (April 22nd, 2026)
2. “Random Walk Models for Population Genetics” *Guest presentation, Advanced Probability Theory IV (Class), Department of Biostatistics, Johns Hopkins University* (April 15th, 2026)
3. “Bayesian Hierarchical Modelling of Population-Level Obesity” *Wearable and Implantable Technology Working Group, Department of Biostatistics, Johns Hopkins University* (January 24th, 2025)

Service and Outreach

Co-lead, Tea-time	Biostatistics Department, Johns Hopkins University	August 2025 – May 2026
PhD Student Mentor	Biostatistics Department, Johns Hopkins University	May 2025 – Present
Co-lead, Curriculum Committee	Biostatistics Student Organization, Johns Hopkins University	May 2025 – Present
• Designed visualization of the PhD core curriculum which was integrated into departmental materials; organized information sessions on PhD qualifying and oral exams; collated student feedback on curriculum and disseminated to faculty		
Site Manager	Opportunity Oxford Residential Program, University of Oxford	September 2020 & September 2021
Ambassador	UNIQ Summer Program, University of Oxford	July – August 2020
Team Guide	International Mathematics Olympiad, Bath	July 2019
Mathematics Subject Representative	Jesus College JCR, University of Oxford	October 2018 – July 2019
Junior Members’ Scholarship Representative	Jesus College JCR, University of Oxford	January – December 2018

Awards

College Prize	Jesus College, University of Oxford	August 2021
• For distinction in Final Honour School of Mathematics (Part C)		
Summer Studentship	Big Data Institute, University of Oxford	July 2021
• Awarded £3000 for summer studentship		
Open Exhibition	Jesus College, University of Oxford	August 2020
• For first-class performance in third-year undergraduate studies		
College Prize	Jesus College, University of Oxford	January 2018
• For excellent performance in collections (internal college examinations)		

References available upon request